origin refers to the URL receiving the request, we call it the *server origin*. Table 3.3 shows the client and server origins for the sample app.

Table 3.3 Client and server origins for the sample app

	URL	Origin
Client	http://localhost:1111/client.html	http://localhost:1111
Server	http://127.0.0.1:9999/api/posts	http://127.0.0.1:9999

There isn't anything inherently special about an origin. It's merely what a browser uses to group content together. Servers use the origin to determine where a request is coming from. Browsers use the origin to define whether a request is same-origin or cross-origin, and exhibits different behavior for each.

SAME-ORIGIN VERSUS CROSS-ORIGIN REQUESTS

With the definition of origin in place, I can provide a more formal definition for same-origin and cross-origin requests. A request is a same-origin request when the client origin and the server origin are exactly the same. Otherwise the request is a cross-origin request.

This distinction between same-origin and cross-origin requests lies at the heart of CORS. When the client initiates the request, the browser extracts the server origin from the URL of the request. It then compares the server origin against the client origin to determine if the request is same-origin or cross-origin. Browsers allow clients to make same-origin requests without any restrictions. But if the request is cross-origin, the browser uses CORS to determine how to handle the request.

In the sample app, the origin of the page at http://127.0.0.1:9999/client.html is http://127.0.0.1:9999. This matches the origin of the API endpoint at http://127.0.0.1:9999/api/posts. Therefore the request is same-origin. On the other hand, the origin for the page at http://localhost:1111/client.html is http://localhost:1111, which doesn't match the origin http://127.0.0.1:9999. Table 3.4 shows example requests along with whether they are same-origin requests.

Table 3.4 Same-origin versus cross-origin requests

Client origin	Server origin	Same-origin request
http://127.0.0.1:9999	http://127.0.0.1:9999	Yes
http://127.0.0.1:9999	https://127.0.0.1:9999	No (different schemes)
http://localhost:1111	http://localhost:9999	No (different ports)
http://localhost:9999/	http://127.0.0.1:9999	No (different hosts)

The last example in table 3.4 might come as a surprise. The IP address for localhost is traditionally 127.0.0.1, so you'd expect http://localhost:9999 and http://127.0.0.1:9999